



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2015
YEAR 11 TASK 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 6-9
- Marks may be deducted for careless or badly arranged work

Total marks – 44

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2 (5 marks)

- Attempt Question 1-5

Section II – Pages 3-5 (39 marks)

- Attempt questions 6-9

Answer all questions in the appropriate space in the Answer booklet provided.

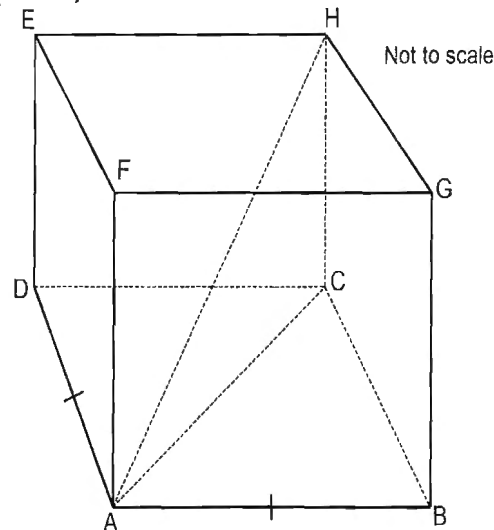
Section I – Multiple Choice - 5 marks

1. Solve for x ,

$$\frac{2x + 1}{1 - x} \geq 1$$

- (A) $0 \leq x < 1$
- (B) $x \leq 0$ or $x > 1$
- (C) $x > 0$ or $x > 1$
- (D) $0 < x \leq 1$

2. A rectangular prism with a square base ABCD, is shown below. The diagonal of the prism, $AH = 8\text{cm}$, the height of the prism, $HC = 4\text{cm}$.



The volume of this rectangular prism is

- (A) 64 cm^3
- (B) 96 cm^3
- (C) 128 cm^3
- (D) 192 cm^3
3. The domain of the function $f(x) = (4 - x^2)^{-\frac{1}{2}}$
- (A) $x \leq -2$ or $x \geq 2$
- (B) $x < -2$ or $x > 2$
- (C) $-2 \leq x \leq 2$
- (D) $-2 < x < 2$
4. If the equation $f(2x) - 2f(x) = 0$ is true for all real values of x , then $f(x)$ could be
- (A) $\frac{x^2}{2}$
- (B) $2x$
- (C) $\sqrt{2x}$
- (D) $x - 2$
5. Ten people are to be seated around a circular table. How many possible seating arrangements are there if two particular friends want to sit directly opposite each other?
- (A) $2 \times 8!$
- (B) $2 \times 9!$
- (C) $4! \times 4!$
- (D) $8!$

End of Section 1

Section II – Extended Response

Attempt questions 6-9.

All necessary working should be shown in every question.

Question 6 (10 marks)

Marks

- a) Prove the identity

2

$$\frac{2 \sin^3 x + 2 \cos^3 x}{\sin^2 x + \sin x \cdot \cos x} = 2 \operatorname{cosec} x - 2 \cos x$$

- b) Determine if the function

2

$f(x) = x^2 + \cos x$ is odd, even or neither. Show all working.

- c) Solve

3

$$\frac{x^2 + 2}{x} \geq 2x - 1$$

- d) . A committee of 5 is to be chosen from 6 men and 8 women.

Find how many committees are possible, if

1

- i) the committee will consist of 3 men and 2 women.

- ii) there is at least one woman on the committee.

2

Question 7 (10marks)

- a) (i) Sketch on the same number plane the graph of $y = \sqrt{3 - x}$ and $y = |x - 1|$.

2

- (ii) Hence or otherwise solve $\sqrt{3 - x} \leq |x - 1|$

2

- b) Solve the equation $3 \cot \theta = \tan \theta + 2$ for $0^\circ \leq \theta \leq 360^\circ$, giving your answer correct to the nearest minute.

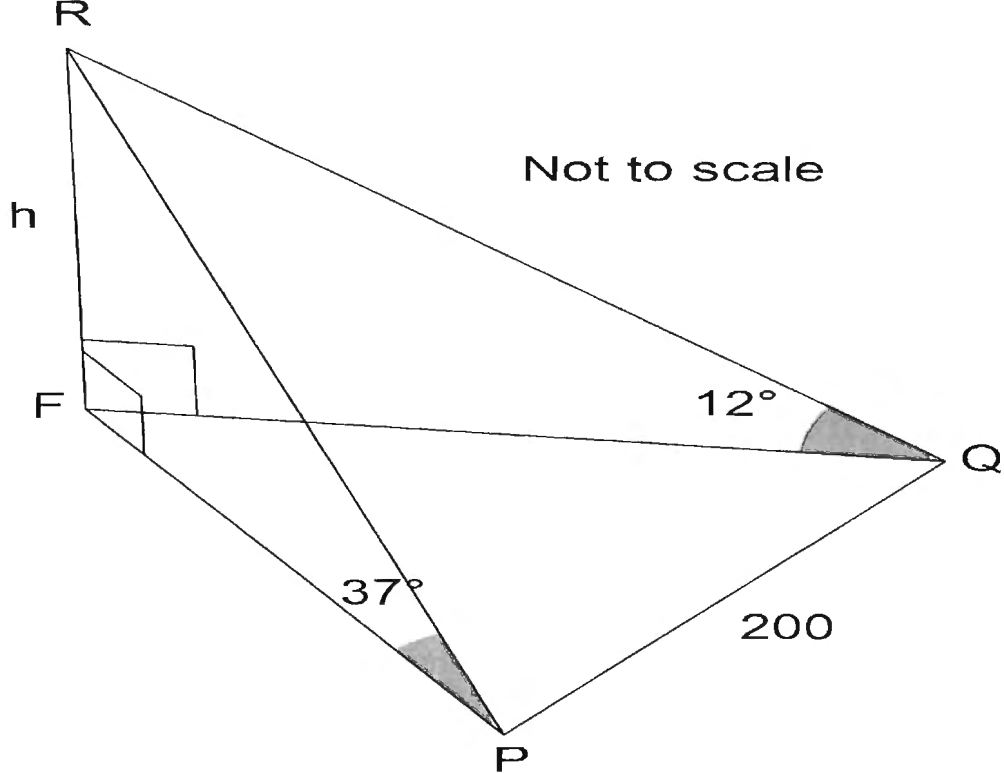
3

- c) Factorise completely $x^5 + x^2 y^2 (y - x) - y^5$

3

Question 8 (8 marks)

- a) A bushwalker walking on a horizontal straight road PQ observes that from his position P the bearing of a hill FR is 337° and he notices the peak R of the hill at an angle of elevation of 37° . After walking 200metres, he arrives at Q. The angle of elevation of R from Q is 12° and Q is due east of the hill.



- i) Show that $FP = h \tan 53^\circ$ 1
- ii) By finding a similar expression for FQ , show that 2
- $$200^2 = h^2 \tan^2 53^\circ + h^2 \tan^2 78^\circ - 2h^2 \tan 53^\circ \tan 78^\circ \cos 67^\circ$$
- 1
- iii) Hence find the height of the tower. 1
- b) , At a dinner party, the host, hostess and their six guests sit at a round table. In how many ways can they be arranged, if the host and hostess are not sitting together? 2
- c) How many different numbers greater than 6000 can be formed with the digits 4, 5, 6, 7, 8 if no digit is used more than once? 2

Question 9 (11 marks)**Marks**

- a) (i) How many different arrangements of the letters of the word ISOSCELES are possible? 2
- (ii) How many of these arrangements have all S's together? 2
- (iii) How many of them have the letter S as the first and last letter? 2
- b) Given a function
$$y = \frac{x}{9 - x^2}$$
- (i) Find all the asymptotes of the function. 2
- (ii) Determine whether the function is even, odd or neither. Justify your answer. 1
- (iii) Sketch the curve. 2

End of Exam

Question 6

a) Prove $\frac{2\sin^3 x + 2\cos^3 x}{\sin^2 x + \sin x \cos x} = 2\operatorname{cosec} x - 2\cos x$

Proof: LHS = $\frac{2\sin^3 x + 2\cos^3 x}{\sin^2 x + \sin x \cos x} = \frac{2(\sin^3 x + \cos^3 x)}{\sin x(\sin x + \cos x)}$

① = $\frac{2(\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)}{\sin x(\sin x + \cos x)}$
= $\frac{2(1 - \sin x \cos x)}{\sin x} = \frac{2 - 2\sin x \cos x}{\sin x}$

① = $\frac{2}{\sin x} - \frac{2\sin x \cos x}{\sin x} = \frac{2}{\sin x} - 2\cos x$

not show
= $2\operatorname{cosec} x - 2\cos x = \text{RHS} \therefore \text{proven}$

b) $f(x) = x^2 + \cos x$

① $f(-x) = (-x)^2 + \cos(-x)$ but $\cos(-x) = \cos x$ even function
= $x^2 + \cos x$
= $f(x) \therefore \text{even function}$

if do $f(-x) = x^2 + \cos x$

but then state $f(x) \neq f(-x)$

and $-f(x) \neq f(-x) \therefore \text{neither} \therefore 1/2$

c) $\frac{x^2+2}{x} \geq 2x-1 \quad x \neq 0$

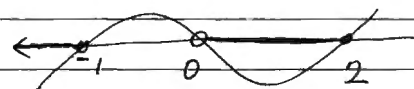
$x(x^2+2) \geq x^2(2x-1)$
 $0 \geq x^2(2x-1) - x(x^2+2)$ ①

$0 \geq x[x(2x-1) - (x^2+2)]$

$0 \geq x[2x^2 - x - x^2 - 2]$

$0 \geq x[x^2 - x - 2]$

$0 \geq x[x-2][x+1]$ ①



$\therefore x \leq -1, 0 < x \leq 2$ ①

d) i) 6 men $\rightarrow$ pick 3 $\therefore {}^6C_3$
8 women $\rightarrow$ pick 2 $\therefore {}^8C_2$

answer ${}^6C_3 \times {}^8C_2 = 560$ ①

ii) At least one women chosen ① for recognising all combinations
 $\therefore 1w \text{ or } 2w \text{ or } 3w \text{ or } 4w \text{ or } 5w$ at least 1 correct

= ${}^8C_1 \times {}^6C_4 + {}^8C_2 \times {}^6C_3 + {}^8C_3 \times {}^6C_2 + {}^8C_4 \times {}^6C_1 + {}^8C_5 = 1996$

OR All possible - No women on committee
= ${}^{14}C_5 - {}^6C_5 \therefore \text{only men}$

= $2002 - 6 = 1996$

if ${}^{14}P_5 - {}^6P_5 = 239520$ 1 mark

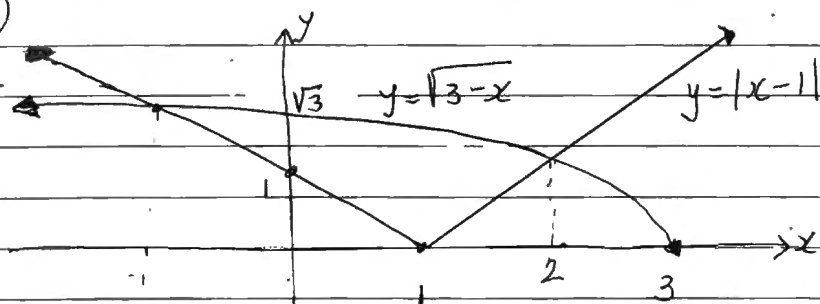
Question 7

a)

$$y = \sqrt{3-x}$$

$$y = |x-1|$$

(i)



① mark for each graph

(pts. of intersection don't need to be shown)

(ii) $\sqrt{3-x} = |x-1|$ first

$$3-x = (x-1)^2$$

$$0 = x^2 - 2x + 1 + x - 3$$

$$0 = x^2 - x - 2 = (x-2)(x+1)$$

$$\therefore x = 2 \quad x = -1 \quad \text{①}$$

$$\therefore \sqrt{3-x} \leq |x-1|$$

① soln: $x \leq -1$ or $2 \leq x \leq 3$

b) $3 \cot \theta = \tan \theta + 2 \quad 0^\circ \leq \theta \leq 360^\circ$

$$\frac{3}{\tan \theta} = \tan \theta + 2 \quad / \times \tan \theta$$

$$3 = \tan^2 \theta + 2 \tan \theta$$

$$0 = \tan^2 \theta + 2 \tan \theta - 3$$

$$0 = (\tan \theta + 3)(\tan \theta - 1)$$

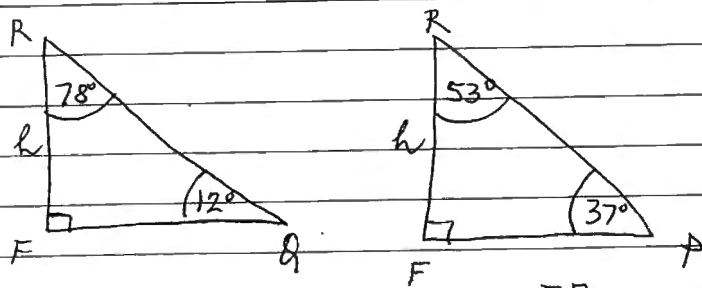
$$\therefore \tan \theta = -3 \text{ or } \tan \theta = 1$$

$$\theta = 108^\circ 26', 288^\circ 26' \quad \text{①} \quad \theta = 45^\circ, 225^\circ$$

c)
$$\begin{aligned} & x^5 + x^2 y^2 (y-x) - y^5 \\ &= x^5 + x^2 y^3 - x^3 y^2 - y^5 \\ &= x^2 (x^3 + y^3) - y^2 (x^3 + y^3) \quad \text{①} \end{aligned}$$

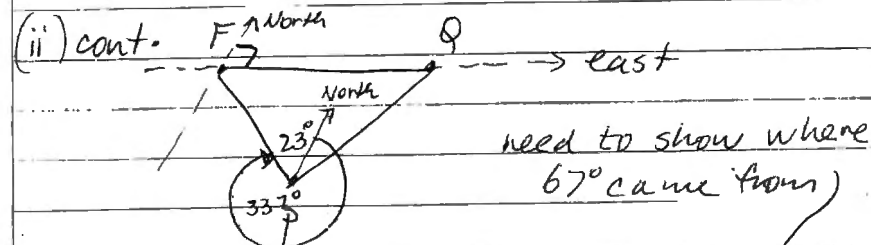
$$\begin{aligned} &= (x^2 - y^2)(x^3 + y^3) = (x-y)(x+y)(x+y)(x^2 - xy + y^2) \\ &\quad \text{①} \quad \text{①} \\ &\text{or } = (x+y)^2 (x-y)(x^2 - xy + y^2) \end{aligned}$$

Question 8



(i) $\tan 53^\circ = \frac{FP}{h}$
 need to show some evidence

need to show $FQ = h \cdot \tan 78^\circ$ $\therefore FP = h \cdot \tan 53^\circ$
 (1)



need to show where 67° came from

$\therefore \angle QFP = 180^\circ - 23^\circ - 90^\circ = 67^\circ$ (1)

$\therefore$ by cosine rule in $\triangle FQP$

$$PQ^2 = FP^2 + FQ^2 - 2 \times FP \times FQ \times \cos 67^\circ$$

$$\therefore 200^2 = h^2 \tan^2 53^\circ + h^2 \tan^2 78^\circ - 2 \times h \tan 53^\circ \times h \tan 78^\circ \times \cos 67^\circ$$

$$\therefore 200^2 = h^2 \tan^2 53^\circ + h^2 \tan^2 78^\circ - 2h^2 \tan 53^\circ \tan 78^\circ \cos 67^\circ$$

$\therefore$ proven

a) iii)

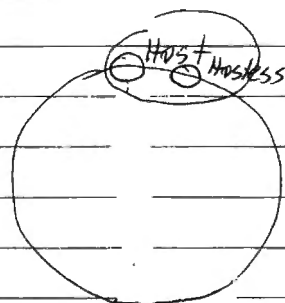
$$200^2 = h^2 [\tan^2 53^\circ + \tan^2 78^\circ - 2 \tan 53^\circ \tan 78^\circ \cos 67^\circ]$$

$$\therefore h^2 = \frac{200^2}{\tan^2 53^\circ + \tan^2 78^\circ - 2 \tan 53^\circ \tan 78^\circ \cos 67^\circ}$$

$$\therefore h = \frac{200}{\sqrt{\tan^2 53^\circ + \tan^2 78^\circ - 2 \tan 53^\circ \tan 78^\circ \cos 67^\circ}}$$

$$\therefore h = 45.864 \text{ (3 d.p.)} \approx 46 \text{ m} \quad \text{[ignore rounding]}$$

b)



Answer = All possibilities - Host Hostess together

$$= (8-1)! - 2 \times (7-1)!$$

$$= 5040 - 1440 = 3600$$

2- marks $\rightarrow$ correct solution with working

1- mark $\rightarrow$ well presented methods with incorrect conclusion

(c) 4 digit numbers - must start with

$\begin{pmatrix} 6 \\ 7 \\ 8 \end{pmatrix}$ ---
 3 more digits

$$\therefore 3 \times {}^4P_3 = 72 \quad \textcircled{1}$$

$$\text{5 digit numbers} = 5! = {}^5P_5 = 120$$

$$\therefore \text{Total} = 72 + 120 = \textcircled{192}$$

2 marks - correct answer with working

1 mark - 4 digits or 5 digit numbers
 or coherent working - mistake obvious

You may ask for extra writing paper if you need more space to answer question

Question 9

(a) (i) $\frac{9!}{3! 2!} = 30240$ $\textcircled{\text{ISOSCELES}}$
 S repeats 3 times $3!$
 E repeats 2 times $2!$

2 marks - correct solution with working
 1 mark = $\frac{9!}{\text{wrong repetition}}$

ii) $\frac{7!}{2! (EE)} = 2520$
 $\textcircled{\text{SSS}}$ any remaining letter 7! 7 places

2 marks - correct solution with working
 1 mark - seeing 7! in the working
 or dividing by 2!

iii) $\frac{7!}{2! (EE)} = 2520$
 S S 7! 7 places
↓ ↓ ↓
Fixed Fixed Fixed

2 marks - correct solution with working
 1 mark - working correctly towards solution

(b) $y = \frac{x}{9-x^2}$

(i) vertical asymptotes $9-x^2 \neq 0$
 $\therefore x = \pm 3$ ①

horizontal asymptote

$x \rightarrow +\infty$ (pick a large number, sub in)

$\therefore y \rightarrow 0^-$

if $x \rightarrow -\infty$ (pick a small number eg. $x = -100$, sub in)

$\therefore y \rightarrow 0^+$

$\therefore$ horizontal asymptote is $y = 0$ ①

ii) $f(x) = y = \frac{x}{9-x^2}$

$f(-x) = \frac{-x}{9-(-x)^2} = \frac{-x}{9-x^2} = -f(x)$ ①

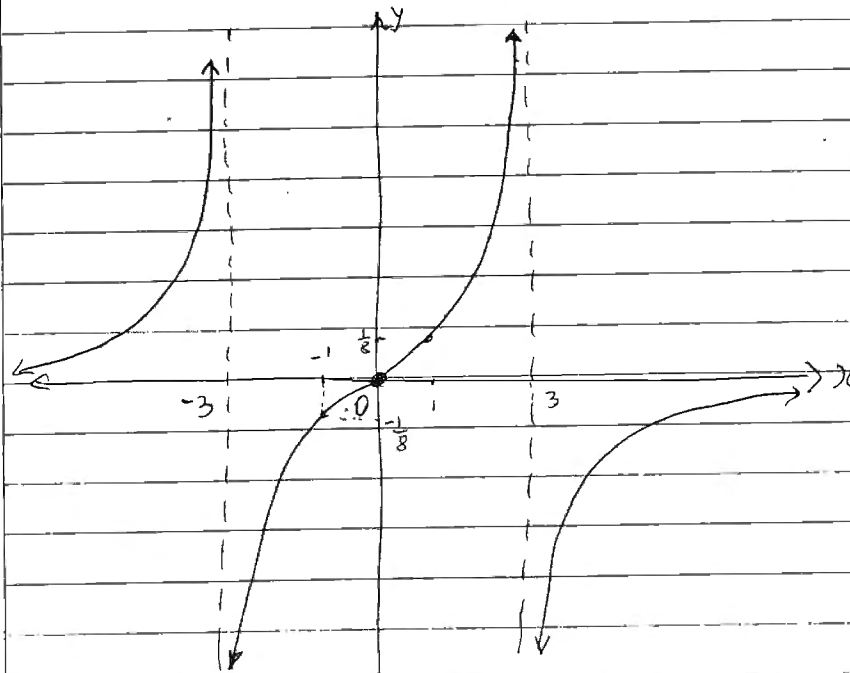
must show
substitution

$\therefore$ odd function

iii)

x	-1	0	1
y	$-\frac{1}{8}$	0	$\frac{1}{8}$

- By using table of values and the property of the odd function drawing the shape



2 marks - correct shape & asymptotes & x-int.

1 mark - showing asymptotes correctly
or odd function features

Multiple choice answers

1. A

2. B

3. D

4. B

5. D